

Vel Tech Group of Institutions

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2028_NeoColab_Competitive Coding II_L8

VTU_Week 12_MCQ

Attempt : 1

Total Mark : 15

Marks Obtained : 11

Section 1 : MCQ

1. The first step in the naive greedy algorithm is _____.

Answer

analysing the zero flow

Status : Correct

Marks : 1/1

2. Suppose you have coins of denominations 1,3 and 4. You use a greedy algorithm, in which you choose the largest denomination coin which is not greater than the remaining sum. For which of the following sums, will the algorithm doesn't produce an optimal answer?

Answer

4

Status : Wrong

Marks : 0/1

3. Which data structure is most commonly used to efficiently implement a greedy algorithm that repeatedly selects the minimum (or maximum) element at each step?

Answer

Heap

Status : Correct

Marks : 1/1

4. What is a greedy algorithm?

Answer

An algorithm that makes the best choice at each step, without considering the overall impact

Status : Correct

Marks : 1/1

5. Which of the following problems is typically solved using a greedy algorithm?

Answer

Coin change problem.

Status : Wrong

Marks : 0/1

6. What is the primary advantage of greedy algorithms over other approaches for certain problems?

Answer

Greedy algorithms do not guarantee the optimal solution

Status : Wrong

Marks : 0/1

7. Which of the following is a disadvantage of greedy algorithm?

Answer

It may not always find the globally optimal solution

Status : Correct

Marks : 1/1

8. Which of the following approaches should be used to find the solution to the activity selection problem?

Answer

Greedy approach

Status : Correct

Marks : 1/1

9. Suppose two activities A and B, have start and finish times as S_A, F_A and S_B, F_B respectively. Both activities are said to be compatible, under which of the following conditions?

Answer

$S_A \leq F_B$ or $S_B \leq F_A$

Status : Correct

Marks : 1/1

10. In the activity selection problem, if two activities have the same finish time, which one is selected?

Answer

The one with the later start time

Status : Wrong

Marks : 0/1

11. Consider the following number of activities with their start and finish time given below. In which sequence will the activity be selected in order to maximize the number of activities, without any conflicts?

Answer

A1, A2, A5

Status : Correct

Marks : 1/1

12. Which of the following is a characteristic of the greedy algorithm?

Answer

It makes the best choice at each step by considering only the local problem state.

Status : Correct

Marks : 1/1

13. What is the primary reason why the greedy algorithm does not work for some optimization problems?

Answer

It does not reconsider previous decisions and makes choices based only on the current state.

Status : Correct

Marks : 1/1

14. In the greedy approach using a stack, what condition do we check before popping from the stack

Answer

If the top is greater than current digit and $k > 0$

Status : Correct

Marks : 1/1

15. Why is sorting not used in this greedy solution?

Answer

All of the mentioned options

Status : Correct

Marks : 1/1